MATH120-24S2 Discrete Mathematics

Assignment 9

Due: 11:59pm Friday 24 October

Submit a single PDF document (1 file only) to the submission link in Learn. It is your responsibility to ensure that your submission is clear and easily readable, and that you upload the correct file by the due date.

1. A graph is k-regular if all of its vertices have degree k.

   (a) Draw a 3-regular graph on 4 vertices.

   (b) Draw a 3-regular graph on 6 vertices.

   (c) Prove that there is no 3-regular graph with 7 vertices.

   Solution:

   (a) For example, K_4 (if the graph is simple, this is the only solution).

   (b) Start with a cycle of 6 vertices, then add three additional edges so that each vertex has degree 3.

   (c) Such a graph would have total sum of degrees $\sum_{v\in V(G)} deg(v) = 3 \times 7 = 21$. By the Handshaking lemma, this is equal to $2|E(G)|$ but 21 is odd which is a contradiction.

2. Determine whether each of the following graphs exists. Provide an example or justify why it does not exist.

   (a) A connected graph G with 10 edges and such that every vertex has the same degree $d > 2$.

   (b) A tree T on 12 vertices with total degree $\sum_{v\in V(T)} deg(v) = 20$.

   Solution:

   (a) E.g. K_5 has five vertices each with degree $d = 4$, and 10 edges. Or you can make a graph with only two vertices, but each of degree 10 (the latter graph is not simple).

   (b) This tree does not exist. A tree on 12 vertices has 11 edges, so the sum of degrees would be $2 \times 11 = 22$ by the Handshaking lemma.

3. Let G be a graph on n vertices with at least $n-1$ edges. Prove that if G contains no cycles then G is connected.
   Hint: start by doing some examples.

   Solution:

   If G contains no cycles then G is a forest. Suppose G has k components G_1,...,G_k, then each component is a tree on $|V(G_i)|-1$ edges and there are a total of $n-k$ edges. However we know $n-k \geq n-1$ and hence $k=1$ and the graph must be connected.